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Model-based geostatistics for mapping disease prevalence from incomplete surveys

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Abstract. National surveys rarely reach every community, yet control programmes need estimates for all of them. We describe a geostatistical model for prevalence data that borrows strength from nearby survey locations and from environmental covariates, and we show how its predictive distribution supports maps of the predicted prevalence, of its uncertainty and of the probability that a threshold is exceeded. In a simulation study the model reduced the mean squared error of predictions by a third compared with a non-spatial alternative. We discuss how to report these results so that they can be used by programme managers.

Keywords: Gaussian process; prevalence mapping; spatial prediction; uncertainty

1 Introduction

Disease control programmes plan their work district by district, but the data that inform those plans come from a limited number of survey locations. Between the locations we must predict, and every prediction carries uncertainty that should travel with it into the decision.

Geostatistical models treat the unobserved prevalence surface as a realisation of a stochastic process. The correlation between two locations decays with the distance between them at a rate set by the scale parameter of the correlation function. Locations close to a survey are predicted with more confidence than locations far from any survey, which is the behaviour a programme needs to see.

In this paper we describe the model, explain how it is fitted and checked, and show how its output can be summarised for decision makers. Section 2 introduces the data and Section 3 the model. Section 4 reports a simulation study, and Section 5 discusses limitations and extensions.

2 Data

The surveys recorded the number of people tested and the number who tested positive in each community, together with the geographical coordinates of the community and a set of environmental covariates derived from satellite images.

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